Note 3 - Transformations of Functions

Transformations

- **Four Shifts**
  - Up by c units: $f(x) + c$
  - Down by c units: $f(x) - c$
  - Right by c units: $f(x - c)$
  - Left by c units: $f(x + c)$
- **Two Reflections**
  - Through x-axis: $-f(x)$
  - Through y-axis: $f(-x)$
- **Vertical Stretch/Shrink**
  - Stretch by a factor of c units: $cf(x)$ where $c > 1$
  - Shrink by a factor of c units: $cf(x)$ where $0 < c < 1$
    Recall: if $c < 0$, then it is a reflection combined with a stretch/shrink!
- **Horizontal Stretch/Shrink**
  - Stretch by a factor of c units: $f(cx)$ where $0 < c < 1$
  - Shrink by a factor of c units: $f(cx)$ where $c > 1$
- **Mother Functions**: Base functions that are often transformed.
  - $y = |x|$
  - $y = \sqrt{x}$
  - $y = x^2$
  - $y = x^3$
- **Order of Operations**: Given a Mother Function, perform transformations in this order:
  1. All stretches or shrinks
  2. All reflections
  3. All shifts

Composite Functions

- f composed with g: $(f \circ g)(x) = f(g(x))$
- g composed with f: $(g \circ f)(x) = g(f(x))$
- f composed with f: $(f \circ f)(x) = f(f(x))$
- g composed with g: $(g \circ g)(x) = g(g(x))$

For composite functions, evaluate the inner function **Method 1:** Fractional (Rational) Functions
Procedure: Find values of x that make the denominator 0.

- Set denominator equal to 0 and solve for x.
- The domain is the set of all real numbers except the values of x found in previous step.
  Recall: A denominator of 0 makes the function undefined.

Example: $h(x) = \dfrac{3-x}{2x+1}$

$$\begin{aligned} 2x+1 &= 0 && \text{Set the denominator equal to 0.} \\ 2x &= -1 && \text{Solve for x.} \\ x &= -\frac{1}{2} \end{aligned}$$

Thus, the domain is everywhere $x \neq -\frac{1}{2}$, since plugging in $-\frac{1}{2}$ would lead to invalid outputs.
In interval notation, this is written as

$$(-\infty, -\frac{1}{2}) \cup (-\frac{1}{2}, \infty)$$

Note: The parentheses are used to show that the interval is not inclusive of $-\frac{1}{2}$ in the domain, but includes numbers greater than $-0.4\overline{999}$ and less than $-0.5000...1$. The $\cup$ represents the union of two subdomains.

Method 2: Even Root Functions *i.e.* $\sqrt{x}, \sqrt[4]{x}, \sqrt[6]{x}...$
Procedure: Find values of x that make the expression inside the root greater than or equal to 0.

- Set the expression inside the radical it greater than or equal to 0, then solve the polynomial inequality.
- The solution to the inequality is the domain.
  Recall: Negative numbers inside even roots lead to imaginary solutions.

Example: $g(x) = \sqrt{x^2 - x - 6}$

$$\begin{aligned} x^2 - x - 6 &\geq 0 && \text{Set the expression greater than or equal to 0.} \\ (x-3)(x+2) &= 0 && \text{Solve for } x. \text{ } \textit{Note: the inequality is temporarily changed to an equality.} \\ x &= 3, -2 && \text{Thus, the inequality is equal to 0 when } x = 3 \text{ or } -2. \end{aligned}$$

When $x = 3$ or -2, then we know that the inequality is equal to 0. However, we must also check whether the inequality is true for the intervals where $x < -2, -2 < x < 3$, and $x > 3$. For each interval, plug in a test point for x and see if the inequality holds true.

- For $x < -2$: plugging in $x = -3$ gives $(-3)^2 + 3 - 6 \geq 0$. When simplified, this gives $6 \geq 0$, which is true. Thus, real numbers less than or equal to -2 are in the domain.
- For $-2 < x < 3$: plugging in $x = 0$ gives $(0)^2 - 0 - 6 \geq 0$. When simplified, this gives $-6 \geq 0$, which is false. Thus, x-values between -2 and 3 are *not* in the domain of this function.
- For $x > 3$: plugging in $x = 4$ gives $(4)^2 - 4 - 6 \geq 0$. When simplified, this gives $6 \geq 0$, which is true.

In interval notation, this is written as

$$(-\infty, -2] \cup [3, \infty)$$

Note: The brackets are used to show that the interval is inclusive of -2 and 3 in the domain.